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Determination of location of multiple harmonic sources in a power system

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Abstract

In this paper, a methodology for accurate determination of multiple harmonic sources in a power system is presented. In the proposed methodology, assuming that no prior information regarding the possible locations of the harmonic sources in the system is available, the problem of determination of location of harmonic sources is solved in two stages. In the first stage, weighted least square estimation technique is used to find possible candidate buses (where the harmonic sources may be connected) among the system buses. In the second stage, using the concept of Euclidean norm, the exact buses (where the harmonic sources are present) among the possible candidate buses are obtained. To demonstrate the effectiveness of the proposed technique, the IEEE-14 and IEEE-30 bus test systems are used.
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1. Introduction

Due to the proliferation of various power electronics based solid-state devices, such as static power converters, static var compensators, cycloconverters, etc. the penetration of harmonics in a power system is ever increasing. As the harmonics are known to cause serious problems in various power system components such as cables, generators, transformer, protective devices, etc. the concern about the assessment and mitigation of power system harmonics is increasing among the electric utilities. Due to this reason, harmonic studies have become an important component of power system analysis and design [1–3]. Harmonic studies are essentially of two types. In the first type, known as the forward harmonics analysis, the harmonic sources in a power system are assumed to be known and harmonic load-flow is carried out to determine the propagation of harmonics in the network. On the other hand, in the reverse harmonic analysis method, the aim is to identify the harmonic sources in a power system from a set of available measurements in the system. The forward harmonic analysis is essentially used to quantify the distortion in the voltage and current waveforms at various points in a power system and also to determine whether dangerous resonant conditions in the power system exist or not.

However, during the last decade or so, interest has been increasing for solving the reverse harmonic problem [4–8]. Heydt [4] first proposed the idea of using least square based state-estimation technique to identify the locations of the harmonic sources. Farach et al. [5] solved the dual problem of optimal sensor location and estimation of location of harmonic sources. A technique for monitoring and identification of harmonic sources by neural network has been proposed in Ref. [6]. In Ref. [7], the problem of identification and tracking of harmonic source has been solved by using the Kaman filter. In Ref. [8], a state-estimation technique has been proposed again to solve the harmonic identification problem. However, in all above approaches, some a priori knowledge of locations of harmonic sources has been assumed for the identification of harmonic sources in the system. Consequently, the important question about the feasibility of identification of harmonic sources in the absence of any a priori knowledge is yet to be addressed.

An attempt has been made here to address the above problem. Towards this goal, a simple two-stage procedure has been proposed to identify the locations of multiple harmonic sources in a power system. Thus the utilities may take subsequent preventive steps in order to mitigate the effects of the harmonics. The paper is organized as follows. In the first stage, probable non-linear (harmonic) buses among the system buses are identified using the weighted least square state-estimation technique and in the second stage, using the concept of Euclidean norm, the exact locations of the harmonic sources are determined.

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2. Solution algorithm

The basic relationship used to determine the locations of the harmonic sources is as follows

$$[I_h] = [Y_h][V_h] \tag{1}$$

In Eq. (1), $[I_h]$, $[Y_h]$, and $[V_h]$ are the bus injection current vector, bus admittance matrix, and bus voltage vector at the hth harmonic order, respectively. Basically, Eq. (1) is the standard relationship connecting the injected bus currents to the bus voltages through the system bus admittance matrix. As the procedure of formation of the above equation is described thoroughly in any power system text book [3], this is not repeated in this paper.

Now, at the buses where the non-linear (harmonic) loads are connected, the value of I_h is non-zero and at the other buses where there are no harmonic loads, the value of I_h is zero. Hence, if the vector $[I_h]$ can be estimated accurately, the harmonic buses would be identified correctly. Moreover, as the bus admittance matrix $[Y_h]$ can be determined accurately from the known system data, the vector $[I_h]$ can be estimated from the knowledge of the vector $[V_h]$. Therefore, the problem of determination of harmonic buses reduces to the problem of estimation of harmonic bus voltages, or, in other words, to the problem of harmonic state-estimation. A brief explanation of the state-estimation technique is given in Appendix A. Based on the above reasoning, the complete algorithm is as follows.

Step 1: This is the first stage of the algorithm. Before carrying out the state-estimation procedure, observability analysis needs to be performed to select proper set of measurements. Observabiltity analysis is carried out with the algorithm [9] forming a completely observable spanning tree. The gain matrix is of full rank thereby establishing numerical observability also. Weighted least square state-estimation technique [9] is then used for the estimation of $[V_h]$ for each harmonic order. After estimating the injection vector $[I_h]$, a set of candidate buses where the harmonic sources may be present are determined. The buses at which the estimated values are close to zero are treated as non-harmonic source buses. Let m be the number of candidate harmonic buses. Obviously $m \leq n$, where n is the number of system buses. Now, out of these m candidate buses, the harmonic source may be present on only any one bus. In this case, the total number of possibilities of harmonic bus is ${}^{m}C_1$. Similarly, the harmonic source may be present on any two of the m candidate buses, thus giving rise to ${}^{m}C_2$ possibilities of harmonic buses. Proceeding as above and collecting all the possibilities, the total number of possibilities of harmonic buses is

$${}^{m}C_1 + {}^{m}C_2 + \cdots + {}^{m}C_m = \sum_{p=1}^{p=m} {}^{m}C_p.$$

To identify the correct harmonic buses out of these total $\sum_{p=1}^{p=m} {}^{m}C_p$ possibilities, the procedure of the second stage is outlined in steps 2 and 3 below.

Step 2: After the current injections at the m buses are obtained, harmonic load-flow studies are carried out for all the above $\sum_{p=1}^{p=m} {}^{m}C_p$ possibilities. The input current injection values required for carrying out the harmonic load-flow studies have been taken as same as the values of the current injections at the m buses obtained from Step 1. After each harmonic load-flow study, the vector of the estimated values of the measurements is obtained. This constitutes the vector of estimated line power and current flows $[\hat{z}]$. Thus a total of $\sum_{p=1}^{p=m} {}^{m}C_p$ vectors are obtained. The procedure for computing $[\hat{z}]$ has been explained in detail in Appendix A and the relevant expression is given in Eq. (A6).

Step 3: For each estimated vector $[\hat{z}]$, the L_2 norm $\|z_m - \hat{z}\|$ is calculated, where $[z_m]$ is the vector of actual measured line power and current flows in the lines. Now, this norm is minimum when $[z_m]$ is closest to $[\hat{z}]$, or, in other words, $[\hat{z}]$ corresponds to the actual combination of harmonic buses in the system. Thus the combination of harmonic buses for which the norm $\|z_m - \hat{z}\|$ is minimum is the actual combination of the harmonic buses present in the system.

In Section 3, the above algorithm is illustrated with the help of numerical results.

3. Numerical validation

To verify the feasibility of the algorithm presented in Section 2, numerical studies have been carried out on both IEEE-14 and IEEE-30 bus test systems. For both test cases the proposed technique has been able to identify the multiple harmonic sources correctly. However, only the results of the IEEE-30 bus system are presented below.

3.1. IEEE-30 bus system

Fig. 1 shows the single line diagram of IEEE-30 system along with the meter locations. The meters have been placed following the observability analysis given in Ref. [9]. The system data are given in Ref. [10]. The following three types of measurements have been considered corresponding to each harmonic order of interest: (a) real power flow, (b) reactive power flow, and (c) magnitude of the feeder current flow. The locations of the meters are shown in Fig. 1 in solid rhombus. In total, there are altogether 29 locations of the actual measurements. In the absence of actual field data, measurements have been simulated by adding a Gaussian noise of zero-mean and known standard deviation to the corresponding true values. The standard deviation has been chosen such that the simulated measurements are within $\pm$ 10% of their corresponding true values. To obtain the true values of the measurements, harmonic load-flow study has been carried out [11]. For this purpose, it has been assumed that the harmonic sources are present at bus numbers 5, 7, and 8. The spectra of the harmonic currents at these buses have been assumed to be the same as that given

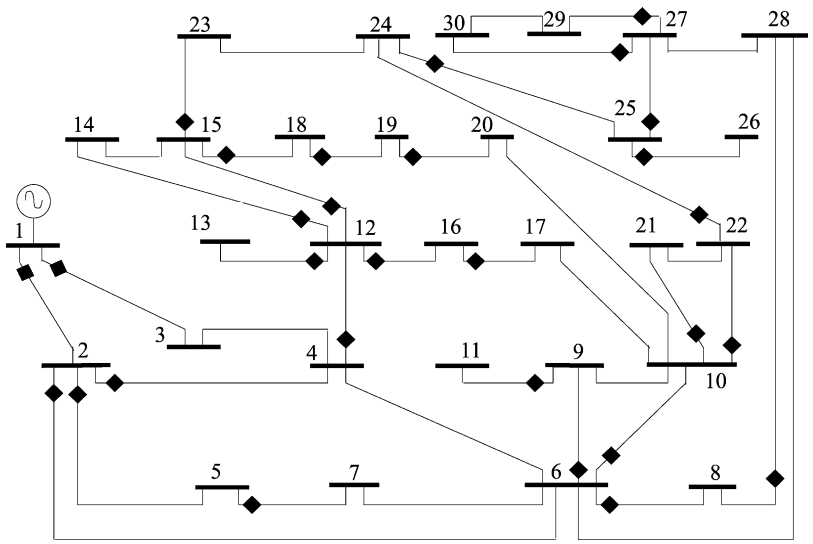


Fig. 1. IEEE 30 bus system. Bus 1 is the slack bus of the system. (◆) P, Q, and I measurements.

in Ref. [12]. It has also been assumed that the spectrum of harmonic voltages at the slack bus, i.e. bus 1 is known from the measurements. Now, it is to be noted that in this case, total number of unknown quantities is 59 for each harmonic order (i.e. the magnitudes of all the buses and angles of all the buses except the slack bus), whereas the total number of actual measurements is 87 (i.e. three types of measurements, viz. real power flow, reactive power flow, and magnitude of the feeder current flow at each of the 29 measurement locations). Thus, the state-estimation problem has sufficient number of redundant measurements, or, in other words, it is a completely over-determined problem to obtain a successful WLS solution. After the values of actual measurements are obtained, harmonic state-estimation has been carried out for each harmonic order and the injection harmonic currents at all the buses are obtained and shown in Table 1.

From Table 1 it is observed that the estimated injected currents are non-zero for buses 5, 7, 8, and 10 (highlighted by boldface letters). For rest of the buses the injected currents are very small and practically zero. Hence, the buses 5, 7, 8, and 10 can be considered as the candidate buses. Thus, there are total $\sum_{p=1}^{p=4} {}^{4}C_{p} = 15$ possibilities of location of different harmonic buses. To determine the actual harmonic buses, L_2 norms of $\|z_m - \hat{z}\|$ for all these 15 different combinations at each harmonic order of interest have been determined. The results are given in Table 2.

From Table 2, it is observed that the L_2 norm is minimum for each harmonic order of interest corresponding to the harmonic loads at buses 5, 7, and 8 (again highlighted by boldface letters). Thus, the estimated harmonic locations are buses 5, 7, and 8, which exactly match with the actual case (the harmonic load-flow has been performed after putting the six pulse rectifier loads at buses 5, 7, and 8). Thus, although bus 10 had been falsely suspected as a possible source of harmonic load by the results of the state-estimator, the calculation of L_2 norm clears the suspicion. Thus, the proposed methodology clearly identifies the multiple harmonic sources in this system. To validate the effectiveness of the proposed method further, the same study has again been carried out when the harmonic sources are assumed to be on buses 2, 5, 12, and 21. To test the robustness of the proposed method for identifying the harmonic sources with

Table 1
Initial estimates with sources taken at buses 5, 7, and 8

Bus no.	Harmonic order					
	5	7	11	13	17	19
1	0.0007	0.0001	0.0007	0.0003	0.0004	0.0005
2	0.0007	0.0007	0.0002	0.0010	0.0005	0.0005
3	0.0009	0.0009	0.0005	0.0009	0.0008	0.0008
4	0.0012	0.0010	0.0012	0.0010	0.0008	0.0001
5	**0.1869**	**0.1252**	**0.0724**	**0.0551**	**0.0361**	**0.0323**
6	0.0017	0.0012	0.0012	0.0014	0.0013	0.0010
7	**0.1277**	**0.0848**	**0.0456**	**0.0342**	**0.0242**	**0.0202**
8	**0.1641**	**0.1089**	**0.0620**	**0.0432**	**0.0292**	**0.0269**
9	0.0016	0.0004	0.0004	0.0005	0.0005	0.0001
10	**0.0156**	**0.0020**	**0.0018**	**0.0031**	**0.0038**	**0.0043**
11	0.0013	0.0003	0.0003	0.0001	0.0001	0.0003
12	0.0012	0.0001	0.0004	0.0001	0.0002	0.0003
13	0.0008	0.0001	0.0000	0.0002	0.0000	0.0001
14	0.0001	0.0000	0.0001	0.0000	0.0001	0.0003
15	0.0004	0.0000	0.0002	0.0003	0.0003	0.0003
16	0.0004	0.0000	0.0002	0.0003	0.0003	0.0001
17	0.0002	0.0004	0.0001	0.0005	0.0004	0.0004
18	0.0003	0.0000	0.0000	0.0004	0.0005	0.0006
19	0.0003	0.0000	0.0007	0.0000	0.0001	0.0001
20	0.0004	0.0006	0.0008	0.0004	0.0004	0.0006
21	0.0009	0.0002	0.0008	0.0003	0.0003	0.0008
22	0.0009	0.0002	0.0004	0.0003	0.0002	0.0008
23	0.0001	0.0004	0.0003	0.0005	0.0005	0.0004
24	0.0002	0.0003	0.0001	0.0004	0.0004	0.0000
25	0.0001	0.0000	0.0001	0.0000	0.0000	0.0000
26	0.0001	0.0000	0.0001	0.0001	0.0001	0.0004
27	0.0008	0.0003	0.0008	0.0003	0.0003	0.0004
28	0.0008	0.0002	0.0000	0.0008	0.0004	0.0002
29	0.0002	0.0000	0.0000	0.0000	0.0000	0.0001
30	0.0002	0.0000	0.0000	0.0000	0.0000	0.0001

Table 2
Calculated L_2 norm with sources taken at buses 5, 7, and 8

Combination of sources	Harmonic order					
	5	7	11	13	17	19
5	0.1056	0.0892	0.0641	0.0434	0.1030	0.0585
7	0.1242	0.0936	0.0689	0.0511	0.0921	0.0915
8	0.0923	0.0816	0.0505	0.0352	0.0559	0.1097
10	0.1791	0.1165	0.0763	0.0624	0.1342	0.1388
5, 7	0.1151	0.0815	0.0559	0.0441	0.0243	0.0462
5, 8	0.0502	0.0229	0.0277	0.0108	0.0243	0.0462
5, 10	0.1116	0.0886	0.0654	0.0442	0.1041	0.0538
7, 8	0.0873	0.0754	0.0470	0.0286	0.0316	0.0918
7, 10	0.1284	0.0934	0.0693	0.0523	0.0860	0.1003
8, 10	0.0914	0.0822	0.0497	0.0348	0.0579	0.1096
5, 7, 8	**0.0017**	**0.00031**	**0.0014**	**0.00013**	**0.00019**	**0.0033**
5, 7, 10	0.1117	0.0828	0.554	0.0445	0.0775	0.0310
5, 8, 10	0.0586	0.0440	0.0216	0.0083	0.0180	0.0461
7, 8, 10	0.0833	0.0739	0.0477	0.0287	0.0428	0.0948
5, 7, 8, 10	0.0049	0.0069	0.0019	0.0018	0.0040	0.0038

different ranges of harmonic injections, the harmonic injections at bus 5 have been taken to be roughly 4, 9, and 5 times higher than the corresponding harmonic injections at buses 2, 12, and 21, respectively. The simulation results are shown in Tables 3 and 4. From the results obtained from these two tables, the presence of harmonic sources at buses 2, 5, 12, and 21 is clearly verified, even though the injected harmonic currents at different buses vary by different order of magnitudes.

It is to be noted that the efficacy of the proposed methodology depends on the quality of the harmonic state-estimation at the first stage. As the quality of the harmonic state-estimation improves, the number of candidate buses obtained at the first stage also reduces. Consequently, the number of different possibilities to be examined in the second stage also reduces. In this work, to obtain a good quality of harmonic state-estimation, measurement set has been so chosen that the system become completely observable and subsequently, an over-determined harmonic state-estimation problem has been solved. However, with ever increasing number of harmonic sources in the system, the number of possible combinations to be checked could still be quite high for a practical size power system, thereby, increasing the computational burden significantly. With the easy availability of modern day fast computers, it would not be a very difficult task to perform these calculations, if necessary. Moreover, as these calculations are to be carried out off-line and once in a while (i.e. not on a frequent regular interval), the need of high computational burden is not going to be a critical hindrance for practical implementation.

4. Conclusions

This paper proposes a new methodology for accurate determination of multiple harmonic sources in a power system. The methodology works perfectly even though there is no a priori knowledge of the presence of the harmonic sources in the network. The methodology solves the problem of identification of harmonic sources

Table 3
Initial estimates with sources taken at buses 2, 5, 12, and 21

Bus no.	Harmonic order					
	5	7	11	13	17	19
1	0.0002	0.0001	0.0000	0.0003	0.0001	0.0004
2	**0.0476**	**0.0319**	**0.0188**	**0.0146**	**0.0094**	**0.0078**
3	0.0006	0.0007	0.0008	0.0007	0.0004	0.0009
4	0.0003	0.0005	0.0010	0.0009	0.0006	0.0010
5	**0.1863**	**0.1251**	**0.0724**	**0.0556**	**0.0359**	**0.0313**
6	0.0007	0.0009	0.0009	0.0002	0.0007	0.0010
7	0.0009	0.0010	0.0007	0.0006	0.0005	0.0001
8	0.0002	0.0005	0.0009	0.0001	0.0001	0.0002
9	0.0005	0.0003	0.0007	0.0004	0.0002	0.0000
10	0.0010	0.0012	0.0018	0.0010	0.0007	0.0003
11	0.0005	0.0003	0.0002	0.0000	0.0000	0.0000
12	**0.0251**	**0.0166**	**0.0094**	**0.0075**	**0.0051**	**0.0042**
13	0.0003	0.0001	0.0001	0.0001	0.0000	0.0000
14	0.0000	0.0001	0.0001	0.0000	0.0000	0.0000
15	0.0001	0.0001	0.0006	0.0001	0.0000	0.0000
16	0.0000	0.0003	0.0004	0.0001	0.0000	0.0000
17	0.0002	0.0009	0.0009	0.0005	0.0000	0.0001
18	0.0001	0.0000	0.0002	0.0009	0.0001	0.0000
19	0.0001	0.0001	0.0000	0.0004	0.0000	0.0000
20	0.0000	0.0002	0.0008	0.0009	0.0000	0.0001
21	**0.0388**	**0.0272**	**0.0157**	**0.0116**	**0.0076**	**0.0068**
22	0.0008	0.0005	0.0002	0.0004	0.0002	0.0001
23	0.0001	0.0002	0.0001	0.0001	0.0000	0.0001
24	0.0001	0.0002	0.0000	0.0001	0.0001	0.0000
25	0.0000	0.0001	0.0001	0.0000	0.0000	0.0000
26	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
27	0.0004	0.0002	0.0000	0.0000	0.0003	0.0003
28	0.0006	0.0003	0.0006	0.0002	0.0002	0.0000
29	0.0001	0.0001	0.0000	0.0000	0.0000	0.0000
30	0.0000	0.0001	0.0000	0.0000	0.0000	0.0000

Table 4
Calculated L_2 norm with sources taken at buses 2, 5, 12, and 21

Combination of sources	Harmonic order					
	5	7	11	13	17	19
2	0.0884	0.0749	0.0554	0.0365	0.0321	0.0453
5	0.0416	0.0261	0.0156	0.0131	0.0079	0.0058
12	0.0974	0.0831	0.0711	0.0447	0.0366	0.0482
21	0.0865	0.0738	0.0685	0.0444	0.0363	0.0484
2, 5	0.0342	0.0244	0.0139	0.0101	0.0053	0.0049
2, 12	0.0844	0.0728	0.0550	0.0363	0.0318	0.0454
2, 21	0.0793	0.0692	0.0504	0.0354	0.0321	0.0448
5, 12	0.0322	0.0243	0.0144	0.0118	0.0078	0.0054
5, 21	0.0151	0.0126	0.0096	0.0086	0.0064	0.0044
12, 21	0.0857	0.0700	0.0679	0.0438	0.0366	0.0478
2, 5, 12	0.0292	0.0216	0.0116	0.0083	0.0049	0.0042
2, 5, 21	0.0113	0.0088	0.0059	0.0042	0.0026	0.0030
2, 12, 21	0.0801	0.0671	0.0505	0.0347	0.0315	0.0439
5, 12, 21	0.0175	0.0098	0.0073	0.0076	0.0060	0.0035
2, 5, 12, 21	**0.0013**	**0.0011**	**3.3×10^{-4}**	**3.3×10^{-4}**	**3.2×10^{-04}**	**2.0×10^{-04}**

in two stages. The possible candidate locations are identified in the first stage and the actual determination of the location of the sources is carried out in the second stage. The algorithm has been tested out on IEEE-14 and IEEE-30 bus test systems. The results obtained clearly show that the developed methodology always identifies the locations of the multiple harmonic sources in the system without any ambiguity.

Appendix A. The state-estimation technique

The state-estimation technique determines the states (the harmonic bus voltages) of the system at the frequency of interest from available measurements and known system configuration. In practice, these variables are estimated from the various measurements (e.g. active and reactive power flow, line current magnitudes, voltage magnitudes, etc.) taken in a power system. The non-linear function [9] of the state-estimation represented in compact form is

$$z_m = h(x) + v \tag{A1}$$

where z_m is the $(m \times 1)$ measurements vector, x is the $(2n - 1)$ state variable vector, $h(\cdot)$ is the $(m \times 1)$ known non-linear function vector, and v is the $(m \times 1)$ errors vector. The system equation, Eq. (A1) can be linearized by a Taylor series expansion of the first order about an initial state x_0 (which is determined from the measurements vector and the system configuration). Thus

$$z_m = h(x_0) + \frac{\partial h(x)}{\partial(x)}(x - x_0) + v \tag{A2}$$

Let $z_0 = h(x_0)$, and $H = \partial h(x)/\partial(x)$; the Jacobian matrix, Eq. (A2) then becomes of the form

$$z_m - z_0 = H(x - x_0) + v \tag{A3}$$

Again setting $\Delta z = z_m - z_0$ and $\Delta x = x - x_0$, Eq. (A3) becomes

$$\Delta z = H\Delta x + v \tag{A4}$$

This linear set of equation is then solved iteratively by (WLS) criteria until convergence is obtained (i.e. $\Delta x < 10^{-5}$). The change in state vector is updated at each iteration k, as

$$x^{k+1} = x^k + \Delta x^k \tag{A5}$$

After the convergence, the vector $[\hat{z}]$ can be determined using estimated state vector $\hat{x}$ as

$$[\hat{z}] = h(\hat{x}) \tag{A6}$$

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